

Android Pilot

A security consultant claims that he has developed an Android App that can remotely hijack airplanes. <http://dgit.in/17sSVLZ>



Turn it off

The new Xbox will have to be connected to the net throughout the day. This article explains why that is a bad idea. <http://dgit.in/XGLGhR>

Space Age



Not so beautiful when it's all you see for four years.

the issues which plague astronauts. There were instances in a Russian cosmonaut program in which crew members refused to speak to each other for months because they had an argument. Though there are interactions with people on the ground, research has shown that sometimes these negative feelings are displaced onto ground teams, going as far as to withhold information from ground personnel in effect giving a very incomplete idea of the occurrences on the spacecraft. This, coupled with the fact that after reaching Mars the astronauts have to wait for one and half years for Mars and the Earth to re-align make the whole endeavor even more depressing. Now astronauts are made to go through a tight regime and sometimes unstable ones slip through, and so what is of concern here is not the danger of psychological problems while travelling in space but in fact the very

same strict regime we mentioned which should be tighter and better planned. Well, even after considering that, there is still the very real problem of radiation poisoning. It has been estimated that an astronaut on the International Space Station experiences radiation tantamount to 8 chest X-rays. In one day. There are also these things called galactic cosmic radiation which are made up of pieces of atoms flung off from supernovas. Due to the Earth's magnetic field, there is some protection from this. After exiting the magnetic field though, it falls with full force. Also, the affects of radiation are accumulative, so the longer you stay exposed to it, even if it is lesser, the accumulation of the radiation will result in greater damage. Because of this radiation, the astronauts can even develop cataracts which is a clouding of the lens inside the eye which can eventually lead to blindness. If that isn't enough, then you should also know that long exposure to conditions such as weightlessness in space can result in a weaker immune system, tissue loss at a rate of 1.5% loss per month, the brain forgetting how to activate the sympathetic nervous system in response to gravity, and the fact that some tests have shown that astronauts (after returning from long bouts in space) have longer reaction times.

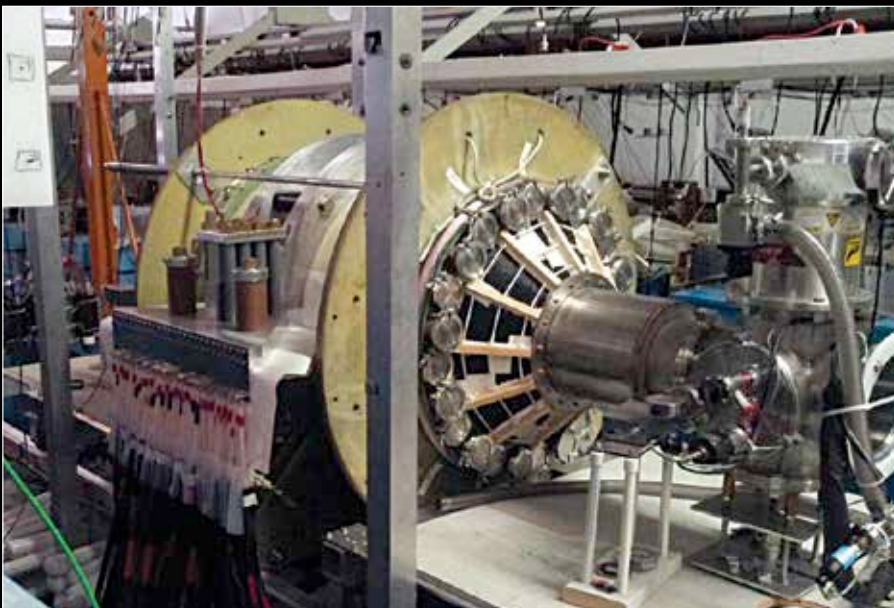
If that isn't enough, consider the logistics. To take a crew to Mars, a lot of chemical fuel is required. Along with this,

supplies are also required to sustain for four years. Because so much fuel is needed, even more fuel is needed because the fuel makes the whole setup even heavier. So that is the situation right now.

And now, we re-introduce the nuclear fusion powered rockets.

Nuclear fusion is a nuclear reaction in which two or more atomic nuclei collide at very high speed and join to form a new type of atomic nucleus. It is the same process that powers active stars. With respect to the rocket, the scientists have created a type of plasma which consists of the hydrogen isotopes- deuterium-tritium. The flow of the plasma through the combustion chamber and towards the exhaust produces a magnetic field. This magnetic field causes the lithium rings to come together and close around this plasma resulting in an implosion. The sheer pressure of this implosion causes the fuel to compress into fusion, which causes the lithium rings to get ejected out of back of the rocket at around 70,000 miles per hour pushing the rocket ahead and hence giving it the required thrust. This process would repeat every ten seconds resulting in the rocket accelerating ahead at around 200,000 miles per hour. The analogy stated by the scientists is that one grain of sand of this new material can generate as much energy as a gallon a rocket fuel resulting in a very efficient system. Now the problem here is not the initiation of the fusion reaction, which is easy enough in itself. The difficulty lies in continuing that reaction. Fusion releases a large amount of thermal energy. To create a fusion reaction even a strong magnetic field is enough, however, to continue the reaction it is imperative that enough of the thermal energy is converted to electrical energy. This is why fusion powered rockets are still theory, however strong a theory it may be. . Currently, using traditional chemical fuel, it would take four years for a round trip to Mars, and the launch in itself will cost a massive 12 billion dollars.

With reduced time, reduced costs, and reduced chances of people killing each other due to the stress of being confined together for years, we assume that astronauts are praying for nuclear fusion powered rockets to work. If they aren't praying, well, they probably should be. 



The fusion driven rocket test chamber where all the magic happens. Theoretically